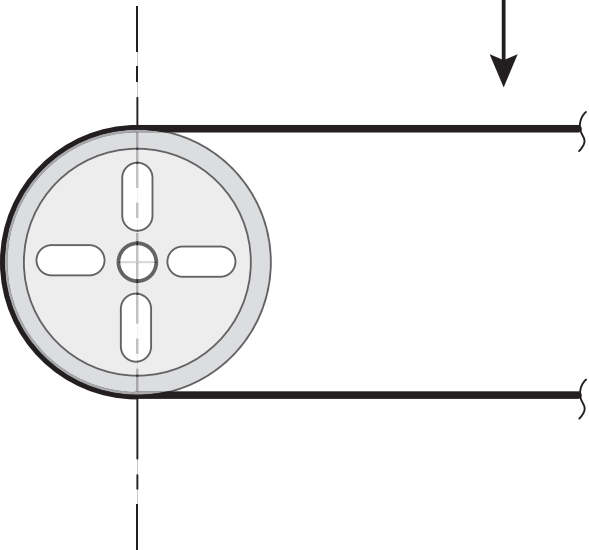
exogenous disturbance

$$d_f(x,t) = F \cdot \delta(x - x_0)\cos(\omega_0 t)$$